Climate Explorer

Historical climate trends with visible assumptions

Be An Earth System Trend Detective!

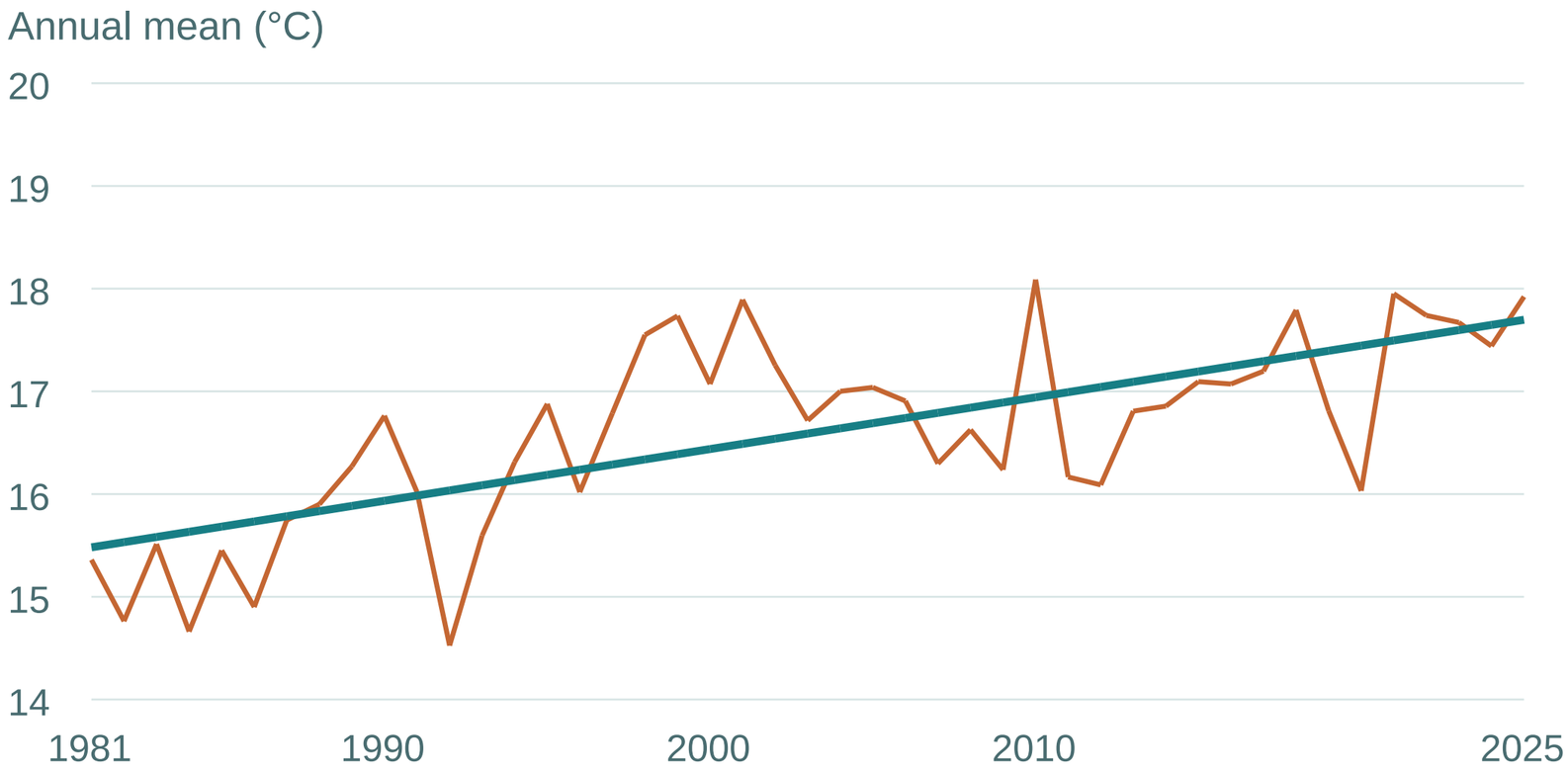
A proposed investigation tool for students and teachers

Work began 11 November 2026, before the event.
Early work and AI ideation require organizer review.

Not affiliated with NASA or the existing KNMI Climate Explorer.

A historical rate is the start of an investigation

Tehran temperature | NASA POWER | 1981-2025 | 45 complete years



+0.504

°C per decade

Fitted historical rate

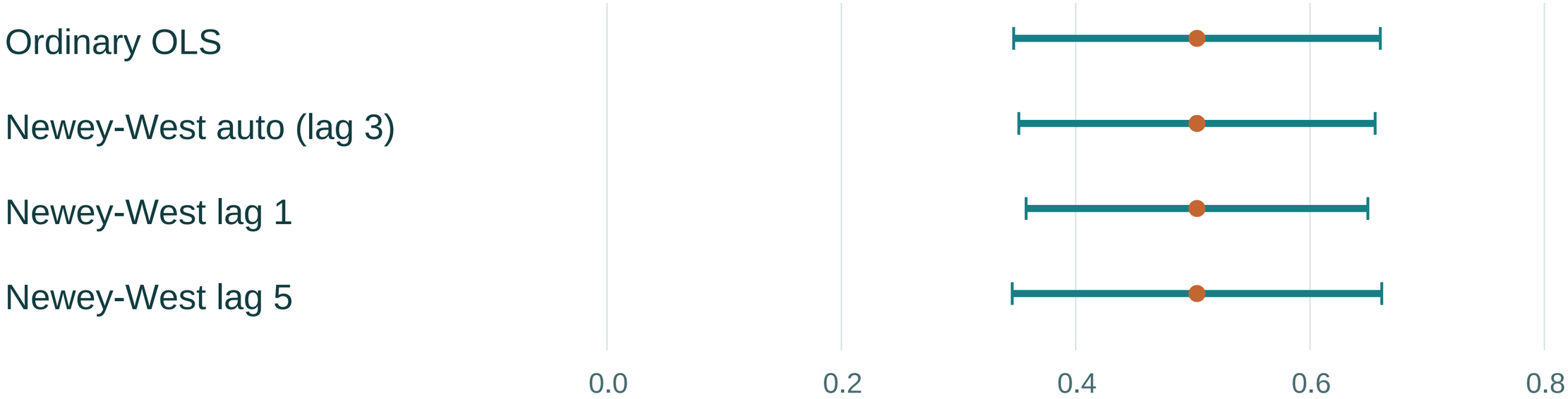
Orange: annual values
Teal: fitted line

A model grid is not a street thermometer. The rate does not identify a cause.

Frozen NASA response and calculation: artifacts/tehran-temperature-v02.json

The same rate can have different uncertainty

Tehran 1981-2025 | Conditional 95% intervals in °C per decade



Changing the dates changes the estimated rate

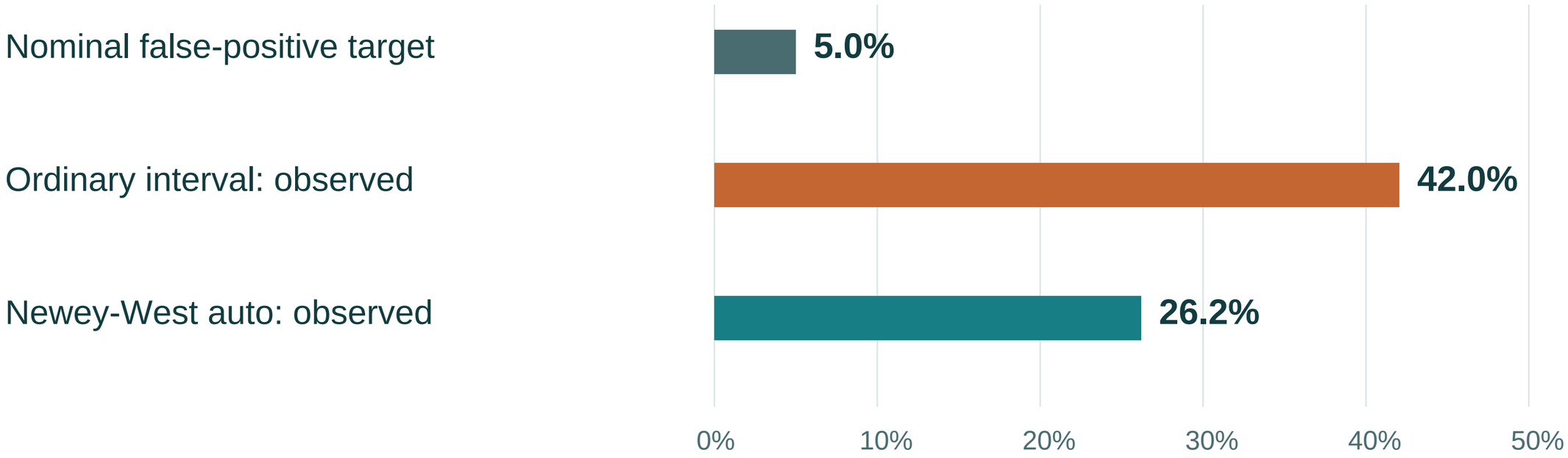
1981-2025: 0.504 | 1986-2020: 0.344 °C per decade

Agreement on shared data does not remove source bias or guarantee coverage.

Intervals are sensitivity checks, not a universal statistical correction.

Some intervals are too certain in our experiment

No true trend | 45 annual values | AR(1) correlation 0.7 | 2,000 records



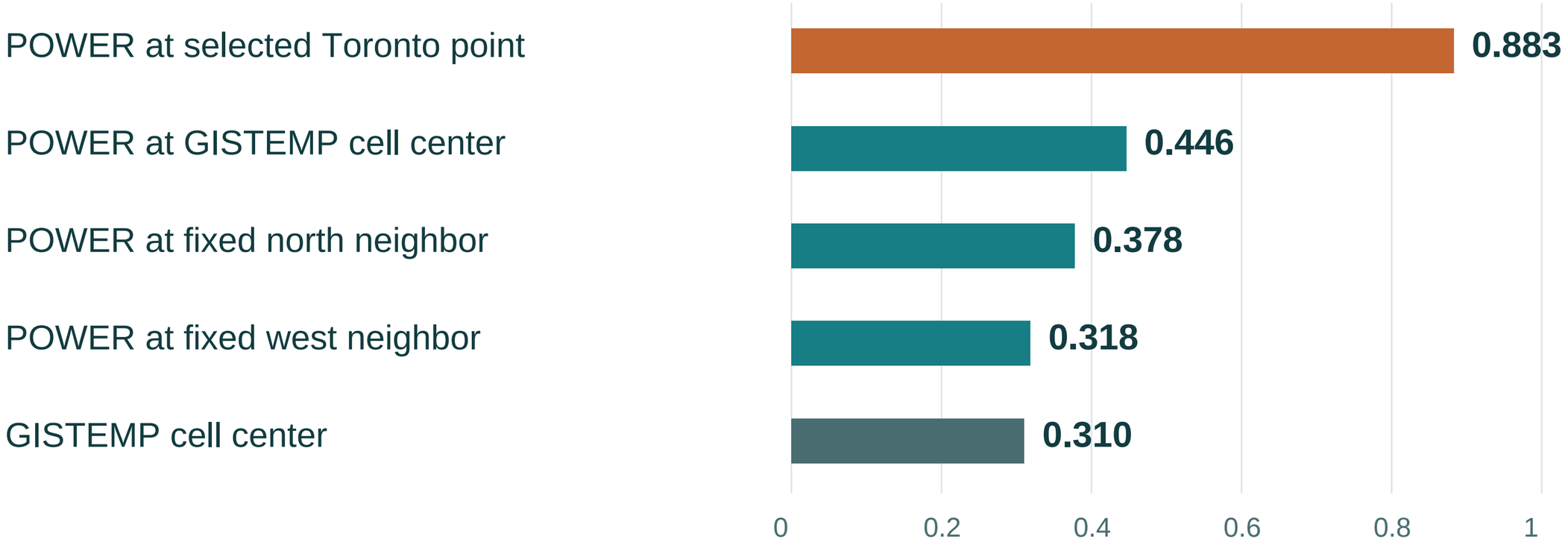
A narrow interval can fail when neighboring years are related.

These rates describe this synthetic scenario, not a real city's error probability.

Full fixed study: 10 scenarios, 20,000 records, 8 method configurations. evaluation/

A source disagreement led to a spatial check

Toronto investigation | Annual temperature rate | 1981-2025 | °C per decade



Changing the POWER location changes the comparison substantially.

Different spatial supports. No station truth or physical cause established.

Fixed exploratory diagnostic: evaluation/SCIENCE_AUDIT.md. All requested points retained.

A result can be recalculated without NASA access

The downloaded record carries raw source data and analysis settings

Saved evidence

Request and retrieval date
Units and complete annual values
Method settings and versions
Source and code hashes

Offline replay

The retained Tehran example replays:
45 annual values, trend and
sensitivity checks agree.
No network is used.

```
python evaluation/replay.py <downloaded-analysis.json>
```

A matching hash detects a change relative to the saved record. It does not prove the data correct.

Replay evidence uses the retained example. Source file SHA-256: e815a85815e3e3eac4ad

Working prototype and next evidence

Public frozen demonstration and reproducible source

Try the frozen examples

https://armandamirchilou.github.io/climate-explorer/

Read the source and evidence

https://github.com/ArmanDamirchilou/climate-explorer

Next: observe real student tasks and improve the explanations.

Educational impact is not yet measured. This tool does not forecast or establish causes.

Codex contributed the approach, code and documentation.

Early work and AI ideation remain unresolved eligibility concerns.

AI_ASSISTANCE.md and submission/REQUIREMENTS.md retain the full history and rules.